

# bzst: Block-Compressed Zstandard

A parallel, seekable, zstd-compatible container format

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*Status of this document: **DRAFT**. This is an in-progress design document, not a ratified specification. Field layouts, magic numbers, and integer widths marked provisional are subject to change; see Section 10.*

**Abstract.** bzst (“beast”<sup>1</sup>) is a container format built on the Zstandard frame format ([RFC 8878](#)). It stores a stream as a sequence of independently-compressed blocks so that compression and decompression can be parallelized, and it embeds a self-contained index that maps uncompressed offsets to compressed offsets so that any position can be reached in  $O(\log n)$ . A bzst file written in the baseline profile is a valid Zstandard archive: `zstd -d` reproduces the original bytes. bzst is a general-purpose compression container; it is deliberately domain-agnostic and defines no semantic (e.g. genomic) indexing or key/value metadata, providing instead a small set of extension points on which derived formats are built.

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<sup>1</sup>The name is “block-compressed zstd.” Said aloud it is *beast*; homage to the X-Men’s Dr. Henry McCoy — the one who is simultaneously the strongest in the room and the one who read the literature first — is entirely intentional.

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# 1 Introduction

## 1.1 Motivation

The BGZF format has served genomics well as the compression layer beneath BAM, VCF/BCF and Tabix-indexed files, but it is showing its age: it is tied to DEFLATE, it caps blocks at 64 KiB (a limit forced by the 48/16-bit virtual offsets of the BAI and CSI indices, not by anything intrinsic), and its indices are external files with all the attendant naming, staleness and object-store problems. Meanwhile Zstandard offers materially better ratio and much higher throughput than DEFLATE, and its *skippable frame* mechanism provides a clean, standard way to carry format metadata inside an otherwise-ordinary compressed stream.

bzst is a Zstandard-based successor to BGZF with four goals:

**Parallelism** Both compression and decompression **MUST** be trivially parallelizable, whether the input is a seekable file or a forward-only stream (a pipe).

**Seekability** A reader **MUST** be able to map an uncompressed byte offset to a compressed location and begin decompressing there, without scanning the whole file.

**Self-containment** The mapping from uncompressed to compressed space **MUST** live inside the file. No sidecar index is required to random-access the compressed container.

**Compatibility** A file in the baseline profile **MUST** remain a valid Zstandard stream that any conformant `zstd` decoder can decompress to the original bytes.

bzst is *not* genomics-specific. Although it grew out of a genomics need and its recommendations (Section 9) speak to genomic use cases, the core format knows nothing about records, coordinates, references, or samples.

## 1.2 Relationship to prior work

bzst draws directly on several existing designs; Section 13 credits them in full. In brief: it uses Zstandard skippable frames the way the `pzstd` and `seekable_format` “contrib” tools do; it adopts the two-layer index idea (uncompressed↔compressed separated from semantic indexing) and the larger-block and self-contained-index goals from J. Bonfield’s **BGZF2** proposal; and it takes lessons on what to fix from the `seekable_format` spec and its `zeekstd` reimplementation.

## 1.3 Design in one picture

A bzst file is a sequence of Zstandard frames. Structural metadata travels in *skippable* frames (which any `zstd` decoder ignores); payload travels in ordinary *data* frames. Each data frame is immediately preceded by a small skippable *block-header frame* that gives the compressed and uncompressed length of that data frame — this is what makes streaming parallel decode possible without parsing frame internals. A skippable *index frame* at the end maps the file for random access.

```
+-----+
| Header frame          bzst skippable (subtype 0x00) |
+-----+
| Block-header frame    bzst skippable (subtype 0x01) | \
| Data frame           zstd data frame                | /  block 0
+-----+
| [ optional derived-format frames: other skippable magics ] |
+-----+
| Block-header frame    bzst skippable (subtype 0x01) | \
| Data frame           zstd data frame                | /  block 1
+-----+
| ...                  |
+-----+
```

```

| [ optional derived index frame: derived format's own magic ]|
+-----+
| Index frame          bzst skippable (subtype 0x02)      |
| ... entries ...                                           |
| [ EOF trailer: index_offset + magic = last bytes of file ]|
+-----+

```

Two invariants make the picture work and are stated normatively later:

1. **Baseline decodability** (Section 7): in the baseline profile every data frame is an ordinary `zstd` frame, so concatenating the decoded output of all data frames — which is exactly what `zstd -d` does, skipping the skippable frames — yields the original uncompressed stream.
2. **Adjacency** (Section 5): a block-header frame and its data frame are physically adjacent, so the two can always be paired.

## 2 Conventions and terminology

The key words **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT**, and **MAY** are to be interpreted as in RFC 2119.

All multi-byte integers are **unsigned** and **little-endian**, matching the Zstandard frame format and existing genomics tooling. `u8`, `u16`, `u32`, `u64` denote unsigned integers of that width.

**CRC32** denotes the CRC-32 computed by `zlib`'s `crc32()` and used by `gzip` (RFC 1952) and `PNG`, catalogued as CRC-32/ISO-HDLC: polynomial `0x04C11DB7` (`0xEDB88320` bit-reflected), initial value `0xFFFFFFFF`, input and output bit-reflected, final XOR `0xFFFFFFFF`. Its check value is `CRC32("123456789") = 0xCBF43926`. Every `bzst` structural-frame checksum is a CRC32, stored as a `u32`.

**Frame** A Zstandard frame as defined by RFC 8878: either a *data frame* (magic `0xFD2FB528`) or a *skippable frame* (magic `0x184D2A50–0x184D2A5F`).

**Skippable frame** A frame carrying an opaque payload that a conformant Zstandard decoder skips. Its wire form is a 4-byte magic, a 4-byte `u32` `Frame_Size`, and `Frame_Size` bytes of user data.

**Data frame** An ordinary Zstandard (or, in a future profile, other-codec) frame carrying compressed payload.

**Block** The pair [block-header frame][data frame] — the unit of independent (de)compression and the unit the index addresses.

**Structural frame** A `bzst`-defined skippable frame (header, block-header, index, ...). Structural frames all share one reserved skippable magic and are distinguished by a subtype byte (Section 3).

**Derived format** A higher-level format (e.g. a `BAM` or `BCF` encoding) layered on `bzst`, which uses its own frames — carried under *other* skippable magics — for its own metadata and indices.

**Profile** A payload-codec capability (Section 7). The *baseline* profile is plain `zstd`, no dictionary; a file declares the set of profiles it uses.

## 3 The bzst skippable-frame namespace

Zstandard reserves sixteen skippable magic numbers, `0x184D2A50` through `0x184D2A5F`. This is a small, globally-shared, uncoordinated space (for example `pzstd` conventionally uses `0x184D2A50` and `seekable_format` uses `0x184D2A5E`). `bzst` therefore claims **exactly one** of these values for all of its structural frames and distinguishes their roles with an internal subtype byte, leaving the other fifteen magics for derived formats and other tools.

**Note.** Provisional magic: `bzst` uses `0x184D2A5B`. Final selection (and the choice to avoid the `0x50` and `0x5E` conventions) is Section 10.2.

Every bzst structural frame has this envelope:

| Bytes        | Field        | Value / Type | Description                                                              |
|--------------|--------------|--------------|--------------------------------------------------------------------------|
| 4            | Magic_Number | 0x184D2A5B   | bzst structural-frame magic (little-endian on disk).                     |
| 4            | Frame_Size   | u32          | Number of bytes that follow (everything below this row).                 |
| 1            | Subtype      | u8           | Structural-frame role; see registry below.                               |
| Frame_Size-1 | Payload      | bytes        | Subtype-specific content, defined per subtype in the following sections. |

The bzst **format version** is a whole-file property, declared once in the header frame (Section 4); it is not repeated per frame. A reader **MUST** skip any structural frame whose Subtype it does not recognize. Because structural frames are skippable, an unknown subtype never impairs decompression — only the higher-level function that subtype would have provided. Each structural-frame payload places its fixed fields at the front and **MAY** carry additional trailing fields in later format versions; readers use the fixed leading fields, locate any trailing checksum via Frame\_Size, and ignore trailing bytes they do not recognize.

#### Subtype registry.

| Subtype   | Name        | Purpose                                                                                         |
|-----------|-------------|-------------------------------------------------------------------------------------------------|
| 0x00      | Header      | File signature, format version, profiles. Exactly one, first frame in the file (Section 4).     |
| 0x01      | BlockHeader | Compressed + uncompressed length of the following data frame, plus per-block flags (Section 5). |
| 0x02      | Index       | Uncompressed→compressed jump table; last frame in the file (Section 6).                         |
| 0x03      | Dictionary  | <i>Reserved.</i> Embedded compression dictionary for the dictionary profile (Section 10.1).     |
| 0x04–0xFF | —           | Reserved for future bzst use.                                                                   |

Derived formats **SHOULD** place their own metadata in skippable frames using one of the other fifteen skippable magics, and **SHOULD** adopt the same (magic, u32 Frame\_Size, ...) envelope shape for consistency. bzst assigns no meaning to those magics and never inspects them.

## 4 Header frame (subtype 0x00)

The header frame **MUST** be the first frame in the file. It makes the file recognizable as bzst from its first bytes, declares the format version, and declares which profiles the file uses.

| Bytes | Field            | Value / Type | Description                                                                                               |
|-------|------------------|--------------|-----------------------------------------------------------------------------------------------------------|
| 4     | Magic_Number     | 0x184D2A5B   | Structural-frame magic.                                                                                   |
| 4     | Frame_Size       | u32          | Bytes following.                                                                                          |
| 1     | Subtype          | 0x00         | Header.                                                                                                   |
| 1     | Format_Version   | 0x01         | bzst format version; governs interpretation of the whole file.                                            |
| 4     | Signature        | "BZST"       | 0x42 0x5A 0x53 0x54. Identifies the container as bzst.                                                    |
| 4     | Format_Signature | u32 / tag    | Opaque 4-byte derived-format tag; 0x00000000 = none (generic bzst). bzst never interprets it (Section 8). |

| Bytes | Field    | Value / Type | Description                                                                 |
|-------|----------|--------------|-----------------------------------------------------------------------------|
| 1     | Profiles | u8           | Bitmask of profiles used anywhere in the file (Section 7). 0x00 = baseline. |
| 1     | Flags    | u8           | Reserved; <b>MUST</b> be 0x00 and <b>MUST</b> be ignored on read.           |
| 4     | Checksum | u32          | CRC32 of all preceding bytes of this frame.                                 |

**Recognition.** A file is bzt if its first four bytes are a Zstandard skippable magic and the four bytes at offset 10 are "BZST". A derived format additionally checks the four bytes at offset 14 against its `Format_Signature`; together these form a fixed 8-byte type magic [BZST][tag] that a tool such as `file/libmagic` can match. The test is cheap, is robust to the leading skippable frame being ignored by generic tools, and does not collide with a plain `zstd` file (whose first bytes are the data-frame magic 0xFD2FB528).

**Note.** `Format_Signature` lets a derived format (say `bzt-bam`) recognize its own files from the header alone, without parsing anything bzt-specific. bzt treats it as opaque: it is an extension point, not domain knowledge. Richer content sniffing and all derived metadata remain the derived format's concern, carried in the derived format's own frames (Section 8).

## 5 Block layout: block-header frame and data frame

Payload is stored as a sequence of **blocks**. Each block is a **block-header frame** immediately followed by a **data frame**.

**Adjacency invariant.** A block-header frame and the data frame it describes **MUST** be physically adjacent. No other frame of any kind — structural, derived-format, or data — **MAY** appear between them. (Derived-format frames **MAY** appear before the block-header frame or after the data frame, never between.)

This invariant lets any reader pair a block-header frame with its data frame by adjacency alone, and lets the index address a whole block with a single offset.

### 5.1 Block-header frame (subtype 0x01)

The block-header frame carries the sizes needed to place and decode the following data frame, plus a flags byte. Its fixed leading fields (below) are stable across format versions; later versions **MAY** append additional per-block metadata before the trailing checksum, which a reader locates via `Frame_Size`.

| Bytes | Field             | Value / Type | Description                                                                                               |
|-------|-------------------|--------------|-----------------------------------------------------------------------------------------------------------|
| 4     | Magic_Number      | 0x184D2A5B   | Structural-frame magic.                                                                                   |
| 4     | Frame_Size        | u32          | Bytes following.                                                                                          |
| 1     | Subtype           | 0x01         | BlockHeader.                                                                                              |
| 4     | Compressed_Size   | u32          | Exact on-disk size of the following data frame.                                                           |
| 4     | Uncompressed_Size | u32          | Size of the following data frame's decoded output.                                                        |
| 1     | Flags             | u8           | Bit 0: Stored (advisory; see Section 5.2). Bits 1–7 reserved (0x0).                                       |
| 4     | Checksum          | u32          | CRC32 over all preceding bytes of this frame; the trailing 4 bytes, located via <code>Frame_Size</code> . |

**Block size limit.** A block's `Uncompressed_Size`, and its on-disk length (block-header frame plus data frame, the index's `Block_Length`), **MUST** each be less than  $2^{32}$  bytes. The frames between two consecutive blocks **MUST** likewise total less than  $2^{32}$  bytes. These bounds let the index store every per-block quantity as a `u32` (Section 6).

**TODO.** Whether the block-header checksum should be mandatory, optional (flag-gated), or dropped in favour of the data frame's own content checksum is Section 10.6.

## 5.2 Data frame

A data frame is a standard Zstandard data frame (RFC 8878), magic `0xFD2FB528`. `bzst` adds no constraints beyond those of the active profile (Section 7). The following is an informative summary so implementors need not consult the RFC for routine work; **RFC 8878 is authoritative**.

A `zstd` frame is `Magic_Number` (4 bytes) followed by a `Frame_Header` (2–14 bytes), one or more `Blocks`, and an optional 4-byte `Content_Checksum`.

- **Frame\_Header.** A 1-byte `Frame_Header_Descriptor` selects which optional fields follow: the `Frame_Content_Size` (the decoded size; 0/1/2/4/8 bytes), a `Single_Segment` flag, a `Content_Checksum` flag, and a `Dictionary_ID` (0/1/2/4 bytes). A `Window_Descriptor` byte (giving the decode window / memory size) is present unless `Single_Segment` is set.
- **Blocks.** Each block is a 3-byte `Block_Header` plus content. The header packs `Last_Block` (1 bit, marks the final block of the frame), `Block_Type` (2 bits), and `Block_Size` (21 bits). `Block_Type` is `Raw` (0, stored verbatim), `RLE` (1, one byte repeated), `Compressed` (2), or `Reserved` (3).
- **Content\_Checksum.** If the descriptor's checksum flag is set, a trailing 4 bytes carry the low 32 bits of the `XXH64` of the decoded content. Writers **SHOULD** enable it so any decoder detects per-block payload corruption.

**Store-only data.** Zstandard has a native uncompressed mode at the block level (`Block_Type = Raw`), the analogue of DEFLATE's level-0 stored blocks; `libzstd` emits it for incompressible input. A data frame whose blocks are all `Raw` is effectively uncompressed yet remains a valid `zstd` frame, so baseline decodability (Section 7) still holds. Whether a frame is stored is determinable **only** by parsing its block headers — there is no frame-level “stored” flag — and equal `Compressed_Size` and `Uncompressed_Size` is **not** a reliable signal (framing overhead usually makes a stored frame slightly larger than its content, and a compressed frame's sizes can coincide). `bzst` therefore offers the advisory `Stored` bit in the block-header `Flags` for writers that want cheap store-only detection; it is a hint only, and the data frame's framing is authoritative.

## 5.3 Parallel decode

**Streaming (forward-only).** A reader loops: read the block-header frame, learn `Compressed_Size` and `Uncompressed_Size`, read exactly `Compressed_Size` bytes for the data frame, and hand the pair (compressed bytes, expected output size) to a worker. No frame internals are parsed to find boundaries, and the output buffer is pre-sized exactly. This is possible on a pipe, with no seeking and no index.

**Seekable.** A reader uses the index (Section 6) to find the block covering a target uncompressed offset, then reads and decodes that block.

**Note.** Because the container learns block boundaries from the block-header frames rather than by parsing data frames, it is **codec-agnostic**: the payload codec can change (Section 7) without changing how blocks are found or the file is indexed. This is the mechanism by which `bzst` is “OpenZL-ready” (Section 7).

## 6 The index (subtype 0x02)

The index is a jump table from uncompressed offsets to compressed locations. It is the last frame in the file. It is an **accelerator**, not a source of truth: the same information is recoverable by a single forward pass over the block-header frames, so a file with a missing or damaged index is still fully decodable and streamable — only  $O(1)$  seeking is lost.

### 6.1 Locating the index

The final bytes of the file form a fixed trailer that lets a reader find the index frame from the end without scanning:

| Bytes | Field        | Value / Type | Description                                             |
|-------|--------------|--------------|---------------------------------------------------------|
| 8     | Index_Offset | u64          | Absolute file offset of the index frame's Magic_Number. |
| 4     | EOF_Magic    | 0x8F92EA5B   | Sentinel; the last four bytes of the file.              |

A reader seeks to EOF - 12, reads these twelve bytes and checks EOF\_Magic; absence of EOF\_Magic signals a truncated or non-bzst file. The index's directory and fixed fields sit immediately before the trailer (Section 6.2), so in practice a reader reads a larger tail in one go and finds them there. These twelve bytes are the tail of the index frame's payload (they are within its Frame\_Size), so a generic zstd decoder still skips the whole frame.

**Note.** Provisional EOF\_Magic 0x8F92EA5B echoes the seekable\_format sentinel family (0x8F92EAB1) while being distinct; final value is Section 10.2. Using an absolute Index\_Offset (rather than a distance-from-EOF) is simplest for a single file but interacts with concatenation (Section 10.4).

### 6.2 Index frame contents

| Bytes  | Field              | Value / Type | Description                                                                                             |
|--------|--------------------|--------------|---------------------------------------------------------------------------------------------------------|
| 4      | Magic_Number       | 0x184D2A5B   | Structural-frame magic.                                                                                 |
| 4      | Frame_Size         | u32          | Bytes following.                                                                                        |
| 1      | Subtype            | 0x02         | Index.                                                                                                  |
| varies | Partitions         | zstd frames  | Partition_Count partitions, in block order and contiguous (Section 6.3).                                |
| varies | Directory          | see below    | 32 bytes per partition, in partition order (Section 6.4).                                               |
| 8      | Entry_Count        | u64          | Number of blocks (= number of entries across all partitions).                                           |
| 8      | Total_Uncompressed | u64          | Sum of all blocks' uncompressed sizes; the end sentinel for search.                                     |
| 8      | Blocks_End         | u64          | File offset just past the last block's data frame (the end of the header frame if there are no blocks). |
| 4      | Partition_Count    | u32          | Number of partitions (= number of directory entries).                                                   |
| 4      | Checksum           | u32          | CRC32 over the bytes from the start of Directory to the end of Partition_Count.                         |
| 8      | Index_Offset       | u64          | Trailer (see above).                                                                                    |
| 4      | EOF_Magic          | 0x8F92EA5B   | Trailer; last bytes of the file.                                                                        |

Everything a reader needs to plan a lookup sits at the end of the frame: the directory and the fixed fields occupy the last  $44 + 32 \times \text{Partition\_Count}$  bytes of the file. A reader that fetches the tail of the file (say its last 64 KiB)

obtains the trailer, the counts and the whole directory in one read, then fetches a single partition per lookup. `Blocks_End` and `Index_Offset` together bracket whatever was written after the last block and before the index, so derived formats can find their trailing frames from the same read (Section 8.1). Checksum covers exactly that contiguous region, so it can be verified without reading any partition; each partition is protected by its own zstd content checksum. The leading envelope fields are validated by value instead (Section 6.5); there is no index flags or version field, because the header's `Format_Version` already governs how the whole file, index included, is interpreted.

Logically the index holds one entry per block, in block order:

**Uncompressed\_Offset** Uncompressed byte offset at which the block's decoded data begins. (Binary-search key.)

**Block\_Offset** Absolute file offset of the block's **block-header frame**.

**Block\_Length** On-disk length of [block-header frame + data frame].

This entry shape is deliberate. A single read of `Block_Length` bytes starting at `Block_Offset` fetches the block-header frame **and** the whole data frame in one I/O — important on high-latency storage systems (e.g. object stores, network filesystems) — and that one buffer already contains the `Uncompressed_Size` needed to pre-size the decode buffer. The entries are not stored in this form; partitions store per-block sizes from which a reader rebuilds them with running sums.

## 6.3 Partitions

The entries are divided, in block order, into **partitions** of consecutive entries. Each partition is stored as a single zstd frame whose decompressed content is exactly  $12 \times \text{Entry\_Count}$  bytes: three columns of `Entry_Count` u32 values, stored one column after another (every value of the first column, then of the second, then of the third).

| Bytes        | Field                              | Value / Type        | Description                                                                                                                                                                                                                         |
|--------------|------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $4 \times n$ | <code>Uncompressed_Size</code>     | <code>u32[n]</code> | Each block's decoded size; non-zero.                                                                                                                                                                                                |
| $4 \times n$ | <code>Derived_Frames_Length</code> | <code>u32[n]</code> | Bytes of other frames (derived-format or any other skippable frames) between the end of the previous block and this block's block-header frame. <b>MUST</b> be 0 for a partition's first entry, whose position the directory gives. |
| $4 \times n$ | <code>Block_Length</code>          | <code>u32[n]</code> | On-disk length of [block-header frame + data frame].                                                                                                                                                                                |

A partition's zstd frame **MUST** set the `Content_Checksum` flag and **MUST** record `Frame_Content_Size`, so a reader can check the decompressed size before allocating for it. A reader rebuilds the logical entries from the partition's directory entry with running sums:

```

Uncompressed_Offset[0] = First_Uncompressed_Offset
Block_Offset[0]        = First_Block_Offset
Uncompressed_Offset[i] = Uncompressed_Offset[i-1] + Uncompressed_Size[i-1]
Block_Offset[i]        = Block_Offset[i-1] + Block_Length[i-1]
                        + Derived_Frames_Length[i]

```

**Note.** Sizes rather than offsets, stored column by column, are what make the index compressible: values within a column cluster around the writer's target block size, and when no other frames sit between blocks the `Derived_Frames_Length` column is all zeros. Measured on BAM, VCF and FASTA files at 64 KiB to 1

MiB blocks, partitions of 4,096 entries compress to about 3–6 bytes per block, against 24 bytes per block for fixed-width absolute entries, while each partition stays independently decodable.

## 6.4 Directory

One 32-byte entry per partition, in partition order:

| Bytes | Field                     | Value / Type | Description                                                              |
|-------|---------------------------|--------------|--------------------------------------------------------------------------|
| 8     | First_Uncompressed_Offset | u64          | Uncompressed_Offset of the partition's first block. (Binary-search key.) |
| 8     | First_Block_Offset        | u64          | Block_Offset of the partition's first block.                             |
| 8     | Partition_Offset          | u64          | Absolute file offset of the partition's zstd frame.                      |
| 4     | Partition_Length          | u32          | On-disk length of the partition's zstd frame.                            |
| 4     | Entry_Count               | u32          | Number of entries (blocks) in the partition; at least 1.                 |

**To seek to uncompressed offset  $x$ :** read the tail of the file, which yields the trailer, the fixed fields and the directory; binary-search the directory for the greatest `First_Uncompressed_Offset`  $\leq x$ ; read `Partition_Length` bytes at `Partition_Offset` and decompress them; walk the running sums to the greatest `Uncompressed_Offset`  $\leq x$ ; read `Block_Length` bytes at `Block_Offset`; decode the data frame; skip  $x - \text{Uncompressed\_Offset}$  bytes into the result. On high-latency storage that is two small reads before the block itself.

## 6.5 Validation

A reader **MUST** treat the index as corrupt, and **MAY** rebuild it by a forward pass over the block-header frames, unless all of the following hold:

- `Magic_Number`, `Subtype` and `EOF_Magic` match; the frame begins at `Index_Offset` and ends at the end of the file; and `Checksum` matches.
- The partitions are contiguous: the first begins immediately after `Subtype`, each begins where the previous one ends, and the last ends where the directory begins.
- Every partition has `Entry_Count`  $\geq 1$ , and the partitions' counts sum to the frame's `Entry_Count`. An empty index has `Partition_Count` = 0 and `Total_Uncompressed` = 0.
- `Blocks_End`  $\leq$  `Index_Offset`, and the last partition's last block ends exactly at `Blocks_End`.
- Each partition declares, and decompresses with a valid content checksum to, exactly  $12 \times \text{Entry\_Count}$  bytes; every `Uncompressed_Size` is non-zero; every `Block_Length` is at least the 22-byte block-header frame; and the first entry's `Derived_Frames_Length` is 0.
- `Entry_Count` is at most  $(\text{Blocks\_End} - 24) / 22$ , the most blocks that fit after the header frame. A reader checks this before decoding any partition, so a crafted, highly compressible partition cannot make the decoded index much larger than the file.
- The first partition's `First_Uncompressed_Offset` is 0; each partition's `Uncompressed_Size` values sum to the next partition's `First_Uncompressed_Offset` minus its own (`Total_Uncompressed` closes the last partition); and each partition's last block ends at or before the next partition's `First_Block_Offset`.

A reader that loads only some partitions checks the rules that involve what it has read.

## 6.6 Partition size

How many entries each partition holds is the writer’s choice, and partitions in one file may differ (the last is usually shorter). A default of 4,096 entries keeps each decompressed partition at 48 KiB. Compression barely changes between about 256 and 4,096 entries per partition, so the choice is about access rather than size: smaller partitions mean less to fetch per lookup, larger ones a smaller directory. Writers of very large files may enlarge partitions to keep the directory within a single modest tail read; a 64 KiB read holds the directory of over 2,000 partitions.

**TODO.** Whether to also store a total compressed length / file length for stronger truncation detection is Section 10.5. An index that exceeds the 4 GiB skippable `Frame_Size` limit (billions of blocks) would need to span multiple frames; deferred (Section 10.4).

## 7 Profiles

A **profile** is a payload-codec capability. A file may use more than one, because whether an individual block uses a dictionary or a non-zstd codec is **self-describing in the data frame itself** — the zstd frame header carries (or omits) a `Dictionary_ID`, and the codec is identified by the data frame’s magic. A reader therefore dispatches per block for free, and a derived format **MAY**, for example, dictionary-compress some blocks and not others with no bzst-level per-block field.

The header’s Profiles byte is a **bitmask of the capabilities used anywhere in the file**, so a reader can fail fast if it lacks one, without scanning every frame:

| Bit | Profile        | Meaning                                                                                                                                                                     |
|-----|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| —   | Baseline       | <code>Profiles == 0x00</code> . Every data frame is a plain zstd frame with no dictionary; zstd -d reproduces the original bytes. <b>MUST</b> be supported by every reader. |
| 0   | Dictionary     | <i>Reserved</i> . Some data frames are zstd-compressed against an embedded dictionary (Section 10.1).                                                                       |
| 1   | Non-zstd codec | <i>Reserved</i> . Some data frames use a non-zstd codec (e.g. a future OpenZL profile).                                                                                     |
| 2–7 | —              | Reserved (0x0).                                                                                                                                                             |

**Baseline requirement.** A generic bzst stream — one produced for direct consumption as .bzst, not as the substrate of a named derived format — **MUST** set `Profiles == 0x00` in v1. This guarantees that any plain .bzst file is decompressible with standard zstd tooling. Derived formats (which identify themselves via `Format_Signature` and their own frames, and are not expected to be consumed by a generic zstd reader) **MAY** use any profile.

**Why profiles, and “OpenZL-ready.”** Two desirable capabilities both break plain-zstd -d decodability: a **dictionary** (a zstd frame that references a dictionary fails to decode without it) and **OpenZL** (whose output is not a zstd frame at all — it has its own container magic 0xD7B1A5C0 and requires the OpenZL universal decoder). Universal decodability therefore cannot be a property of **every** bzst file if these are to be supported; it is instead a guaranteed property of the **baseline** profile. Because the container finds blocks via block-header frames rather than by parsing payload (Section 5), adding a codec later requires **no container change**: a reader dispatches per block on the data frame’s own magic. bzst does not specify the non-zstd codec profile now — OpenZL’s wire format is explicitly pre-1.0 and unstable — but reserves the bit. See Section 12.

## 8 Derived formats

bzst provides the compression container; a derived format provides meaning. The contract is:

- A derived format **MAY** place any number of skippable frames under magics other than bzst's, subject only to the adjacency invariant (Section 5). bzst never inspects them. It **MAY** declare a 4-byte `Format_Signature` in the header (Section 4) for detection.
- A derived format that is record-based **SHOULD NOT** split a record across a block boundary; each block should contain a whole number of records. (bzst itself has no notion of a record; this is guidance for derived formats — see Section 9.)
- Semantic indexing — by genomic coordinate, by record number, by name — is a derived-format concern, layered **on top of** the bzst index. The natural design maps semantic keys to **uncompressed** offsets; the bzst index then maps those to compressed locations. This is the two-layer split that decouples semantic addressing from block size and removes BGZF's 64 KiB ceiling.
- bzst defines **no** generic key/value store, and no JSON/YAML-like structure. A derived format that needs such things defines them in its own frames.

### 8.1 Anchoring a derived index on the bzst index

A derived format that wants its own trailing index (e.g. a genomic coordinate index) places its frames after the last block, where the bzst writer then appends the index. A reader finds them from the same read of the file's tail that it uses for the bzst index:

1. Read the tail of the file (Section 6.2): the trailer gives `Index_Offset` and the index's fixed fields give `Blocks_End`.
2. Everything in `[Blocks_End, Index_Offset)` is what the derived format wrote after the last block and before the bzst index — for example its coordinate index frame(s). The two offsets are equal when nothing was written there.

A reader can then go either way. **Forward:** skippable frames are self-delimiting forward (magic + `Frame_Size`), so it reads the region from `Blocks_End`. **Backward:** frames cannot be walked backward, so a derived format that wants to jump straight to a payload **SHOULD** end its last frame with a fixed-size footer of its own (e.g. the payload's absolute offset followed by an identifying magic), which a reader fetches from just before `Index_Offset`. bzst writes `Blocks_End` itself and never reads the region, so derived formats need no bzst support. This is a concrete demonstration of the two-layer split: the derived (semantic) index and the bzst (compression) index coexist, each self-contained, with bzst unaware of the former.

## 9 Recommendations (non-normative)

**Block size** Choose block size for the access pattern. Smaller blocks (down to BGZF's 64 KiB) give finer random-access granularity and, at  $\leq 64$  KiB **uncompressed**, remain compatible with the 16-bit within-block field of BAI/CSI virtual offsets. Larger blocks (hundreds of KiB to a few MiB) give better ratio and throughput. There is no format-imposed minimum; the maximum is just under 4 GiB (Section 5), far beyond any size useful for random access.

**Record alignment** For record-based data (BAM, BCF, FASTQ), align blocks to record boundaries so no record straddles a block. bzst's generous block-size limit (just under 4 GiB) means even a single very large record (a long read, a wide multi-sample VCF row) can occupy its own block — something BGZF's 64 KiB limit made impossible.

**Checksums** Enable Zstandard's content checksum on data frames.

**Heterogeneous data** For data whose statistics drift along the file (e.g. concatenated FASTQs, name-sorted or multi-sample data), prefer larger blocks, which capture local context per block. A single static dictionary trained on the head of such a file degrades as the data diverges from its training sample (Section 10.1).

**Small block sizes** If small blocks are required (e.g. for BAI/CSI compatibility), a dictionary was explored as a way to offset the resulting ratio cost, but did not prove worthwhile for v1 (Section 10.1).

**Derived-format detection** Declare a `Format_Signature` in the header and, if needed, put a richer identification frame (under the format's own magic) early in the file.

## 10 Open issues

Tracked design questions. Resolved decisions and their rationale are in Section 11.

### 10.1 Dictionary profile

The dictionary subtype (0x03) is reserved but not specified in v1. Initial experiments applying dictionaries to both short-read and long-read data did not show gains sufficient to warrant a dictionary profile in v1. The idea is not ruled out — dictionaries may be revisited in a future version if they prove their worth.

### 10.2 Magic-number selection

Final values for the bzt structural-frame magic (provisional 0x184D2A5B) and the EOF\_Magic (provisional 0x8F92EA5B), avoiding the 0x184D2A50 (pztst) and 0x184D2A5E (seekable\_format) conventions. Open.

### 10.3 Per-block codec tagging

Per-block dictionary/codec use is self-describing (data-frame `Dictionary_ID` and magic), and the header Profiles bitmask enables fast-fail, so no per-block bzt profile field is needed. Remaining question: is dispatch on the data-frame magic sufficient, or is an explicit per-block codec tag (e.g. in the block-header Flags) wanted for robustness? **Lean:** magic dispatch is enough. Open (minor).

### 10.4 Concatenation and oversized indices

Concatenating two complete bzt files yields a valid zstd stream that `zstd -d` decodes correctly but that has two headers and two indices. Define whether tools treat this as one logical stream (re-index) or reject it. Related: an index too large for one 4 GiB skippable frame would need to span multiple frames. Both deferred. Open.

### 10.5 Truncation detection

Whether the index should additionally store a total compressed length / expected file length so that mid-file truncation (not just a missing trailer) is detectable. **Lean:** yes. Open.

### 10.6 Block-header checksum

Whether the per-block block-header checksum should be mandatory (as drafted), flag-gated optional, or dropped in favour of the data frame's own content checksum. A corrupt block-header mis-dispatches a parallel decode, which argues for keeping it. Open.

## 11 Resolved decisions and rationale

- **Scope.** bzt is a generic compression container only — no genomics, no generic key/value or JSON/YAML metadata; derived formats own all semantics via their own frames. *Why:* keeps the base format domain-agnostic and small; semantic indexing is inherently format-specific and belongs a layer up.
- **Universal decodability is a profile property**, guaranteed for the baseline profile, not for every file. A direct `.bzt MUST set Profiles == 0` in v1. *Why:* both dictionaries and OpenZL break plain-zstd -d; scoping universality to a profile preserves the interop promise where it matters.
- **Profiles are a header bitmask of capabilities used; per-block behaviour is self-describing** (data-frame `Dictionary_ID` / magic), so blocks may mix dictionary/non-dictionary and codecs with no per-block bzt field. *Why:* zstd already encodes this per frame; a per-block bzt profile would be redundant.

- **Format version is a whole-file property** declared once in the header, not repeated per frame. *Why:* the header is always read first, so per-frame version bought only self-description-in-isolation, which is unnecessary — and it cost a byte per block.
- **Format\_Signature.** The header carries a 4-byte opaque derived-format tag so a derived format is detectable from a fixed 8-byte [BZST][tag] magic without parsing anything else; bzst never interprets it. *Why:* cheap, standard type detection; keeps derived detection out of derived-only frames while leaving all real metadata to the derived layer.
- **One reserved skippable magic, subtyped internally;** the other fifteen are left to derived formats. Unknown subtypes **MUST** be skipped. *Why:* frugal with the scarce 16-value global magic space; fully bzst-controlled subtype space; no bzst-vs-derived collisions inside a file.
- **BlockHeader (subtype 0x01) rather than “sizing”,** and forward-extensible: fixed leading fields, optional trailing per-block metadata in later versions, trailing checksum located via `Frame_Size`. *Why:* clearer name; the block header is the natural home for future per-block metadata.
- **Advisory Stored flag** for zstd Raw (store-only) data frames; the zstd framing is authoritative and such frames stay baseline-decodable. *Why:* equal sizes are an unreliable signal, and there is no frame-level stored flag in zstd.
- **Native index, not reused seekable\_format.** *Why:* reuse would force offsets reconstructible only by summing from the start of the file, and a dummy entry for every interleaved skippable frame; the interop value (one niche contrib tool) is low.
- **A partitioned, zstd-compressed index with a fixed-width directory at the end,** and no other index encoding. Partitions store per-block sizes as u32 columns; the directory anchors each partition with absolute u64 offsets. *Why:* a lookup reads the file’s tail and one small partition (two reads on high-latency storage) without loading the whole index; on the genomic data measured the index is about 4–7× smaller than fixed 24-byte entries; and implementers need nothing beyond the zstd they already have, little-endian integers and running sums — no variable-length integer codec.
- **The index addresses the block-header frame;** each logical entry is [`uncompressed_offset`, `block_offset`, `block_length`]. *Why:* one read fetches block header + data together and carries the uncompressed size needed to pre-size the buffer, with no size duplicated except the block length.
- **Inline block-header frames are the source of truth;** the index is a reconstructible accelerator. A missing/damaged index never breaks the file.
- **The index records Blocks\_End,** which with `Index_Offset` brackets the region after the last block where derived formats put their own trailing frames (Section 8.1). *Why:* one tail read then locates both the bzst index and any derived index, and the writer knows the value itself; no derived-format value passes through bzst.
- **Little-endian throughout; CRC32 for structural checksums;** recommend zstd content checksums on data frames. *Why:* CRC32 ships with zlib and with the standard libraries of many languages, whereas libzstd does not expose its XXH64 through its public API, so XXH64 would force a separate xxHash dependency on most implementations; speed is irrelevant at our scale.
- **u32 block sizes.** Block-header sizes, and every per-block quantity in the index, are u32, so a block is limited to just under 4 GiB uncompressed and on disk. *Why:* a 4 GiB block is useless for random access, u32 matches `seekable_format/zeekstd` precedent, and it halves the size fields in every block header. Whole-file offsets stay u64.
- **Block size is the writer’s choice** within that limit; no minimum is imposed.
- **No record straddles a block** — a recommendation at the bzst level, a **SHOULD NOT** for record-based derived formats (softened from **MUST NOT**: unforeseen use cases may need otherwise).

## 12 OpenZL: findings and forward-compatibility

OpenZL (Meta, 2025–) is a graph-based, format-aware compressor by the zstd authors. Findings relevant to bzst, from its paper (arXiv 2510.03203), source, and the genomics discussion in issue #76:

- **Its output is not a zstd frame.** OpenZL frames carry their own magic (0xD7B1A5C0-derived) and require the OpenZL universal decoder; even the “fall back to zstd” path stores a **magicless** zstd block inside an OpenZL frame. There is no mode that emits a standard zstd frame. Hence an OpenZL-payload file cannot be a baseline bzst file.
- **Its win is real on ratio and on compression CPU, but not on decode.** Versus high zstd/xz it typically gives 1.3–2× the ratio while compressing several-fold faster, but decompression is often **slower** than zstd for parse-heavy formats (e.g. BAM  $\approx$  249 vs 526 MB/s) — a caveat for a format whose thesis is fast parallel decode.
- **On genomics it beats BAM/BGZF/zstd but trails CRAM  $\approx$  2×,** and trained compressors overfit their training data. So even with an OpenZL profile, bzst is a BGZF/BAM successor and a superior container, not a CRAM-class ratio competitor.
- **Maturity:** v0.x, wire format explicitly changing, backward-decode guaranteed only “for at least the next several years.”

Consequently bzst reserves the non-zstd-codec profile bit but does not specify it now. Forward-compatibility is achieved structurally: the container is codec-agnostic (Section 5), so when OpenZL stabilizes an OpenZL profile can be added with no change to the header, block-header, or index frames.

## 13 Acknowledgements

This design builds directly on: the Zstandard format and its pzstd and seekable\_format contrib tools (Y. Collet, N. Terrell, and the zstd project); the **BGZF2** proposal and prototype (J. Bonfield, Wellcome Sanger Institute), from which the larger-block, self-contained-index, two-layer-index and record-boundary goals are taken; the zeekstd reimplementation (R. Rosen), whose fixes to seekable\_format informed the index design; and OpenZL (Meta), which motivated the codec-agnostic, profile-based structure. BGZF itself (H. Li *et al.*) is the format bzst hopes to succeed.

## 14 References

- RFC 8878 — Zstandard Compression and the application/zstd Media Type. <https://www.rfc-editor.org/rfc/rfc8878>
- Zstandard project and compression-format doc. <https://github.com/facebook/zstd>
- zstd seekable\_format spec. [https://github.com/facebook/zstd/blob/dev/contrib/seekable\\_format/zstd\\_seekable\\_compression\\_format.md](https://github.com/facebook/zstd/blob/dev/contrib/seekable_format/zstd_seekable_compression_format.md)
- zeekstd seekable format. <https://github.com/roosen/zeekstd>
- BGZF2 proposal (J. Bonfield). <https://github.com/jkbonfield/htslib/blob/bgzf2/BGZF2.md>
- OpenZL. <https://github.com/facebook/openzl> and paper arXiv:2510.03203. <https://arxiv.org/abs/2510.03203>
- RFC 1952 — GZIP File Format Specification version 4.3 (defines CRC-32, with sample code). <https://www.rfc-editor.org/rfc/rfc1952>

## 15 Appendix A — Reference tooling wishlist (informative)

A living, non-normative list of tools a reference implementation may want. This is implementation guidance, not part of the format.

- **bzst compress / bzst decompress** — parallel block-wise (de)compression to/from a stream, with a target block size and profile.

- **bzst cat** — concatenate several bzst files into one, recomputing a single jump table (and reconciling headers/profiles) rather than leaving multiple indices (Section 10.4).
- **bzst index** — (re)build or repair the index frame from a forward pass over the block-header frames; validate an existing index against the blocks.
- **bzst info** — dump the header (version, Format\_Signature, profiles), block count, total sizes, per-block sizes, and checksum status.
- **bzst extract** — random-access extraction of an uncompressed byte range via the index.
- **bzst verify** — check structural and (where present) data-frame checksums, the EOF trailer, and truncation.
- **bzst rehydrate** — transcode an advanced-profile file (e.g. dictionary) down to the baseline profile for consumption by generic zstd tooling.
- **bgzf2bzst** / **bzst2bgzf** — convert to/from BGZF for migration.